OS & SP

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SKILL – 11

1.Skill11.c

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GNU nano 8.7.1 skill11.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_HISTORY 10

void trim(char *str)
{
    int start = 0;
    int end = (int)strlen(str) - 1;

    while (str[start] == ' ' || str[start] == '\t')
        start++;

    while (end >= start &&
           (str[end] == ' ' || str[end] == '\t'))
        end--;

    memmove(str, str + start, end - start + 1);
    str[end - start + 1] = '\0';
}

int main(void)
{
    char *history[MAX_HISTORY];
    int count = 0;
    char command[256];

    while (1) {
        printf("ossp11$ ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

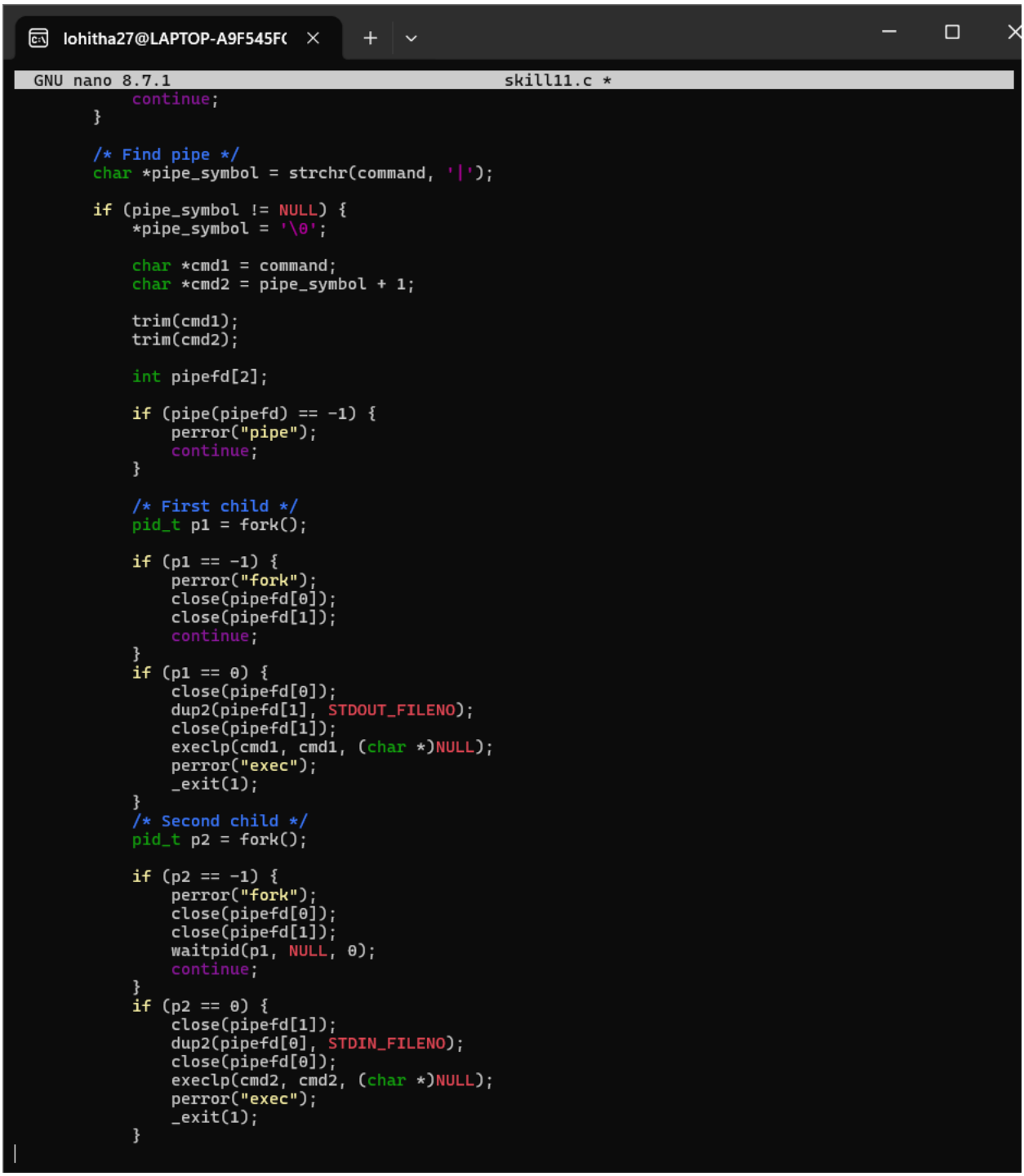
        if (strcmp(command, "exit") == 0)
            break;

        /* Store command */
        if (count < MAX_HISTORY) {
            history[count] = strdup(command);

            if (history[count] == NULL) {
                perror("strdup");
                break;
            }

            count++;
        }

        /* Display history */
        if (strcmp(command, "history") == 0) {
            for (int i = 0; i < count; i++)
                printf("%d %s\n", i + 1, history[i]);
```


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continue;
}
/* Find pipe */
char *pipe_symbol = strchr(command, '|');
if (pipe_symbol != NULL) {
*pipe_symbol = '\0';
char *cmd1 = command;
char *cmd2 = pipe_symbol + 1;
trim(cmd1);
trim(cmd2);
int pipefd[2];
if (pipe(pipefd) == -1) {
perror("pipe");
continue;
}
/* First child */
pid_t p1 = fork();
if (p1 == -1) {
perror("fork");
close(pipefd[0]);
close(pipefd[1]);
continue;
}
if (p1 == 0) {
close(pipefd[0]);
dup2(pipefd[1], STDOUT_FILENO);
close(pipefd[1]);
execlp(cmd1, cmd1, (char *)NULL);
perror("exec");
_exit(1);
}
/* Second child */
pid_t p2 = fork();
if (p2 == -1) {
perror("fork");
close(pipefd[0]);
close(pipefd[1]);
waitpid(p1, NULL, 0);
continue;
}
if (p2 == 0) {
close(pipefd[1]);
dup2(pipefd[0], STDIN_FILENO);
close(pipefd[0]);
execlp(cmd2, cmd2, (char *)NULL);
perror("exec");
_exit(1);
}

```
                waitpid(p1, NULL, 0);
                continue;
            }
            if (p2 == 0) {
                close(pipefd[1]);
                dup2(pipefd[0], STDIN_FILENO);
                close(pipefd[0]);
                execlp(cmd2, cmd2, (char *)NULL);
                perror("exec");
                _exit(1);
            }

            close(pipefd[0]);
            close(pipefd[1]);
            waitpid(p1, NULL, 0);
            waitpid(p2, NULL, 0);
        } else {
            printf("Command stored: %s\n", command);
        }
    }
    for (int i = 0; i < count; i++)
        free(history[i]);
    return 0;
}
```

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill11.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra  -g skill11.c -o skill11
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill11
ossp11$ ls
Command stored: ls
ossp11$ pwd
Command stored: pwd
ossp11$ history
1 ls
2 pwd
3 history
ossp11$
```